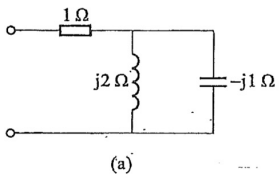
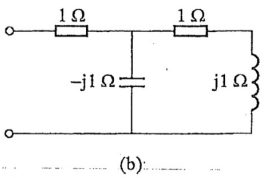
9-1 试求题 9-1 图所示各电路的输入阻抗 Z 和导纳 Y。

(a) $Z = 1 + j2 // (-j1) = 1 - 2j\ \Omega$

$Y = \frac{1}{Z} = \frac{1}{5} + \frac{2}{5}j\ S$



(a)



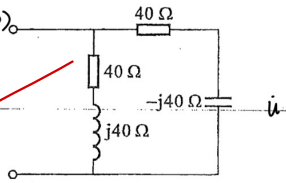
(b)

(b) $Z = 1 + (-j1) // (1 + j1) = 2 - j\ \Omega$

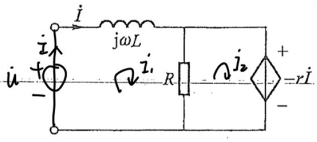
$Y = \frac{1}{Z} = \frac{2}{5} + \frac{1}{5}j\ S$

(c) $Z = (40 + j40) // (40 - j40) = 40\ \Omega$

$Y = \frac{1}{Z} = \frac{1}{40}\ S$



(c)



(d)

(d) $(j\omega L + R)\dot{I}_1 - R\dot{I}_2 = \dot{U}$

$R(\dot{I}_1 - \dot{I}_2) = -r\dot{I}$

$\dot{I}_1 = \dot{I}$

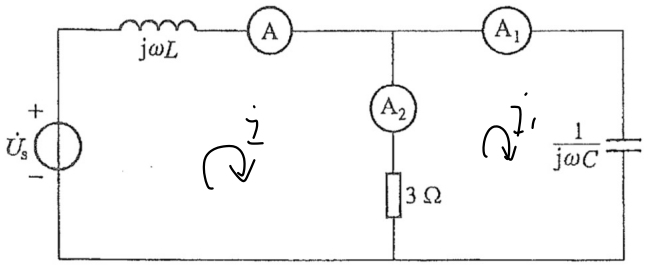
$\Rightarrow \dot{U} = (j\omega L - r)\dot{I}$

$Z = j\omega L - r\ \Omega$

$Y = \frac{1}{Z} = \frac{1}{j\omega L - r}\ S$

题 9-1 图

9-4 已知题 9-4 图所示电路中 $u_S = 16\sqrt{2}\sin(\omega t + 30°)$ V，电流表 A 的读数为 5 A。$\omega L = 4\ \Omega$，求电流表 A_1、A_2 的读数。



题 9-4 图

$u_S = 16\sqrt{2}\cos(\omega t - 60°)$

$\dot{U}_S = 16\angle -60°$

设 $\dot{U}_{3\Omega} = U\angle\varphi$

$\Rightarrow \dot{U}_C = U\angle\varphi$

$\dot{I}_1 = I_1\angle\varphi - 90°$

$\dot{I}_2 = I_2\angle\varphi$

$\dot{I} = \dot{I}_1 + \dot{I}_2 = I_1\angle\varphi - 90° + I_2\angle\varphi = 5\angle\varphi$

$3\dot{I}_2 - \dot{U}_S + \dot{I}\,j\omega L = 0$

$\Rightarrow \begin{cases} 5\angle\varphi = I_1\angle(\varphi - 90°) + I_2\angle\varphi \\ 16\angle -60° = 3I_2\angle\varphi + 4I_1\angle\varphi \end{cases}$

$\begin{cases} 5\cos\varphi = I_1\sin\varphi + I_2\cos\varphi \\ 5\sin\varphi = -I_1\cos\varphi + I_2\sin\varphi \\ 16\cos(-60°) = 3I_2\cos\varphi + 4I_1\cos\varphi - 8 \Rightarrow \cos\varphi = \frac{-8I_2}{3I_2 + 4I_1} \\ 16\sin(-60°) = 3I_2\sin\varphi + 4I_1\sin\varphi = -8\sqrt{3} \Rightarrow \sin\varphi = \frac{-8\sqrt{3}}{3I_2 + 4I_1} \end{cases}$

$\frac{-8I_2}{3I_2 + 4I_1} + \frac{-8\sqrt{3}I_1}{3I_2 + 4I_1} = 5\cos\varphi$

$\frac{-8\sqrt{3}I_2}{3I_2 + 4I_1} + \frac{-8I_1}{3I_2 + 4I_1} = 5\sin\varphi$

$\frac{64(I_2 + \sqrt{3}I_1)^2}{(3I_2 + 4I_1)^2} + \frac{64(I_1 - \sqrt{3}I_2)^2}{(3I_2 + 4I_1)^2} = 25$

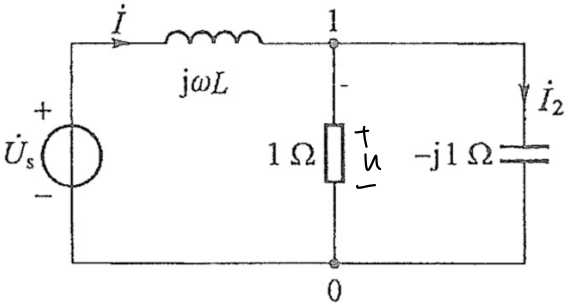
$I_1 = 3A$, $I_2 = 4A$ or $I_1 = 4.799A$, $I_2 = 1.404A$

同除 $\angle\varphi \Rightarrow \begin{cases} 5\angle\varphi - \varphi = I_1\angle -90° + I_2\angle 0 \\ 16\angle -60° - \varphi = 3I_2\angle 0 + 4I_1\angle 0 \end{cases}$

$\because \varphi_1 = \varphi' - \varphi$, $\varphi_2 = -60° - \varphi$

$\Rightarrow \begin{cases} 5\angle\varphi_1 = I_1\angle -90° + I_2\angle 0 \\ 16\angle\varphi_2 = 3I_2\angle 0 + 4I_1\angle 0 \end{cases} \Rightarrow \begin{cases} 5\cos\varphi_1 = I_2 \\ 16\cos\varphi_2 = 3I_2 + 4I_1 \\ 5\sin\varphi_1 = -I_1 \\ 16\sin\varphi_2 = 0 \end{cases}$

9-5 题 9-5 图所示电路中，$I_2 = 10$ A，$U_s = \frac{10}{\sqrt{2}}$ V，求电流 $\dot{I}$ 和电压 $\dot{U}_s$，并画出电路的相量图。



题 9-5 图

设 $\dot{U} = U\angle 0$

$\dot{U}_C = U\angle 0$

$\dot{I}_2 = \frac{\dot{U}}{-j1} = Uj \Rightarrow U = 10V$

$\dot{I}_1 = \frac{\dot{U}}{1}$

$\dot{I} = \dot{I}_1 + \dot{I}_2$

$= U(1 + j) = 10 + j10$ A

$\dot{I}\,j\omega L + \dot{U} = \dot{U}_S$

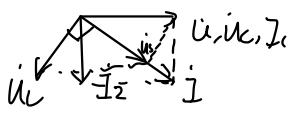
$\dot{U}_S = \omega L(-10 + j10) + 10$

$= -10\omega L + 10 + j10\omega L$

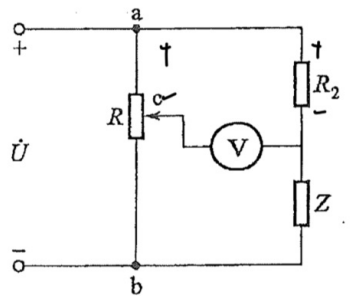
$\Rightarrow (-10\omega L + 10)^2 + (10\omega L)^2 = \left(\frac{10}{\sqrt{2}}\right)^2$

$\Rightarrow \omega L = 0.5$

$\Rightarrow \dot{U}_S = -5 + 10 + j5 = 5 + j5$ V



9-15 在题 9-15 图所示电路中，已知 $U=100$ V，$R_2=6.5\ \Omega$，$R=20\ \Omega$，当调节触点 c 使 $R_{ac}=4\ \Omega$ 时，电压表的读数最小，其值为 30 V。求阻抗 Z。



题 9-15 图

设 $\dot{U}=U\angle 0=100$ V

$\dot{U}_{R_2}=\frac{R_2}{R_2+Z}\dot{U}$

$\dot{U}_{ac}=\frac{4}{R}\cdot\dot{U}$

$\dot{U}_0=\left|\frac{R_2}{R_2+Z}-\frac{4}{R}\right|\dot{U}$

$=\left|\frac{6.5}{6.5+Z}-0.2\right|\cdot 100$

$=30\angle\varphi$

设 $\frac{a}{6.5+Z}=a+jb$ 模小 $\Rightarrow (a-x)^2+b^2$

$a^2-2ax+x^2+b^2$ 在 $x=0.2$ 时模小

$x^2-2ax+a^2+b^2=0.2=\frac{2a}{2}\Rightarrow a=0.2$

$\Rightarrow |b|\cdot j\cdot 100=30\angle\varphi\Rightarrow b=\pm 0.3$

$\frac{6.5}{6.5+Z}=0.2\pm j0.3$

$\Rightarrow Z=3.5\pm j1.5\ \Omega$

9-17 列出题 9-17 图所示电路的回路电流方程和节点电压方程。已知 $u_S=14.14\cos(2t)$ V，$i_S=1.414\cos(2t+30°)$ A。

$\dot{U}_S=10\angle 0$ V　　$\dot{I}_S=1\angle 30°$ A

(a)

$(j5+1)\dot{I}_1-j5\dot{I}_2-\dot{I}_3=-\dot{U}_0$

$-j5\dot{I}_1+(j5-j5)\dot{I}_2+j5\dot{I}_3=\dot{U}_S$

$\dot{I}_3=g\dot{U}_0$

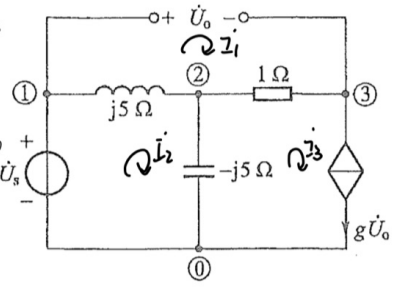
$\dot{I}_1=0$ ✗

$\dot{U}_1=\dot{U}_S$

$-\frac{1}{j5}\dot{U}_1+(\frac{1}{j5}+1-\frac{1}{j5})\dot{U}_2-\dot{U}_3=0$

$-\dot{U}_2+\dot{U}_3=-g\dot{U}_0$

$\dot{U}_0=\dot{U}_1-\dot{U}_3$



(a)

(b)

$(j8+2)\dot{I}_1-(j8+0)\dot{I}_2-\dot{I}_3=\dot{U}_S$

$-(j8+0)\dot{I}_1+(j8+3)\dot{I}_2-\dot{I}_3=0$

$(\dot{I}_3-\dot{I}_2)\cdot 1+(\dot{I}_3-\dot{I}_1)\cdot 1+\dot{I}_4\cdot(-j8)=0$

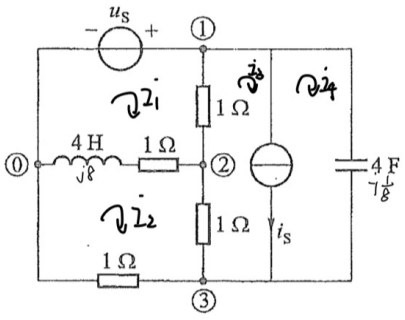
$\dot{I}_3-\dot{I}_4=\dot{I}_S$

$\dot{U}_1=\dot{U}_S$

$-\dot{U}_1+(1+\frac{1}{1+j8}+1)\dot{U}_2-\dot{U}_3=0$

$j\frac{1}{8}\dot{U}_1-\dot{U}_2+2\dot{U}_3=\dot{I}_S$

$(2-j\frac{1}{8})$



(b)

(c)

$(1+1)\dot{I}_1-\dot{I}_2=-\dot{U}_C+2\dot{I}$

$-\dot{I}_1+(1+1-j10)\dot{I}_2+j10\dot{I}_3=0$

$j10\dot{I}_2+(j10-j10)\dot{I}_3=-2\dot{I}$

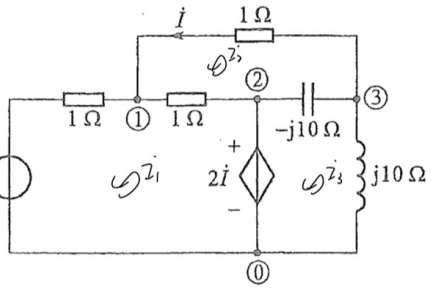
$\dot{I}=\dot{I}_2$

$(1+1+1)\dot{U}_1-\dot{U}_2=\frac{\dot{U}_S}{1}$

$\dot{U}_2=2\dot{I}$

$\frac{1}{j10}\dot{U}_2+(1+\frac{1}{-j10}+\frac{1}{j10})\dot{U}_3=0$

$\dot{I}=\frac{\dot{U}_3-\dot{U}_1}{1}$



(c)

$-\dot{I}_1+(1+2)\dot{I}_2-2\dot{I}_3=0$

$-\dot{U}_S+\dot{U}_1\cdot 1+(\dot{I}_1-\dot{I}_2)\cdot 1+(\dot{I}_3-\dot{I}_2)\cdot 2+\dot{I}_3\cdot j2\cdot 2=0$

$\dot{I}_1-\dot{I}_3=\beta\dot{I}$

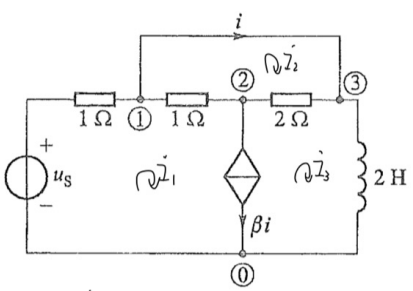
$\dot{I}=\dot{I}_2$

$(1+1)\dot{U}_1-\dot{U}_2=\frac{\dot{U}_S}{1}$

$-\dot{U}_1+(1+2)\dot{U}_2-2\dot{U}_3=-\beta\dot{I}$

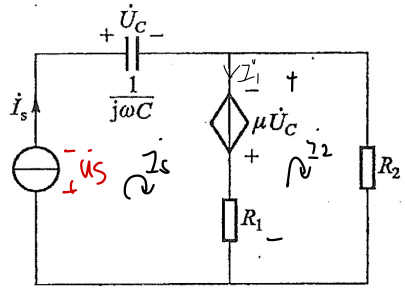
$-2\dot{U}_2+(2+j4)\dot{U}_3=0$

$\dot{U}_1=\dot{U}_3$



(d)

9-18 已知题 9-18 图所示电路中，$I_s=10$ A，$\omega=5\ 000$ rad/s，$R_1=R_2=10\ \Omega$，$C=10\ \mu$F，$\mu=0.5$。求电源发出的复功率。



题 9-18 图

设 $\dot{I}_S=I_S\angle 0=10$A

$-R_1\dot{I}_S+(R_1+R_2)\dot{I}_2=-\mu\dot{U}_C$

$\dot{U}_C=\dot{I}_S\times\frac{1}{j\omega C}=\frac{10}{j\cdot\frac{1}{200}}=\frac{200}{j}$ V　　$U_C=200$V

$\Rightarrow \dot{I}_2=5+j5$ A　　$I_2=5\sqrt{2}$A

$\dot{I}_1=\dot{I}_S-\dot{I}_2=5-j5$ A　　$I_1=5\sqrt{2}$A

$P=I_1^2R_1+I_2^2R_2=1000$W

$Q=U_C^2\cdot\omega C\cdot\cos(-90°)=-2000$ var

$\bar{S}=1000-2000j$ VA ✗题意：源

$\dot{U}_S=\dot{U}_C+\dot{U}_3=-j200+\dot{I}_2R_2=50-j150$ V

$\bar{S}=\dot{U}_S\cdot\dot{I}_S^*=500-j1500$ VA

9-22 题 9-22 图所示电路中 $R_1=R_2=10\ \Omega$，电压表的读数为 20 V，功率表的读数为 120 W。试求 $\dot{U}_2/\dot{U}_s$ 和电源发出的复功率 $\bar{S}$（$L=0.25$ H，$C=10^{-3}$ F）。

$P_2=\frac{U_2^2}{R}=40W$ $P=P_1+P_2=120W$

$\Rightarrow P_1=80W=\frac{U_1^2}{R_1}\Rightarrow U_1=20\sqrt{2}V$ $I_1=2\sqrt{2}A$

设 $\dot{I}_1=2\sqrt{2}\angle\varphi$ $\dot{I}_C=\dot{U}_C\,j\omega C=0.02\omega j\ A$

$\dot{I}_1=\dot{I}_2+\dot{I}_C\Rightarrow 2\sqrt{2}\angle\varphi=2+0.02\omega j$

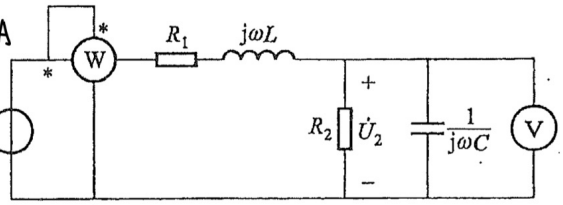
$\Rightarrow 2^2+(0.02\omega)^2=(2\sqrt{2})^2\Rightarrow\omega=100\,rad/s$

$\dot{I}_1=2+j2$ $\Rightarrow\frac{\dot{U}_2}{\dot{U}_S}=-\frac{1}{2\sqrt{5}}-j\frac{2}{2\sqrt{5}}$

$\dot{U}_S=(R_1+j\omega L)\dot{I}_1+\dot{U}_2$

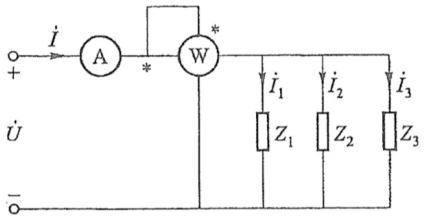
$=-10+70j$

$\bar{S}=\dot{U}_S\dot{I}_1^*=120+j160\ VA$



题 9-22 图

9-25 把 3 个负载并联接到 220 V 正弦电源上，各负载取用的功率和电流分别为：$P_1=4.4$ kW，$I_1=44.7$ A（感性）；$P_2=8.8$ kW，$I_2=50$ A（感性）；$P_3=6.6$ kW，$I_3=60$ A（容性）。求题 9-25 图中表 A、W 的读数和电路的功率因数。



题 9-25 图

设 $\dot{U}=220\angle0°$ V

$P_1=UI_1\cos\alpha$ $P_2=UI_2\cos\beta$ $P_3=UI_3\cos\gamma$

$\Rightarrow\cos\alpha=\frac{200}{447}$ $\Rightarrow\cos\beta=\frac{4}{5}$ $\Rightarrow\cos\gamma=0.5$

$\alpha=63.42°$ $\Rightarrow\beta=36.9°$ $\Rightarrow\gamma=-60°$

$\dot{I}=\dot{I}_1+\dot{I}_2+\dot{I}_3=I_1\angle0+I_2\angle\beta+I_3\angle\gamma=91.77\angle-11.33°$

$\Rightarrow$ A表读数：91.77 A

W表读数 $P=UI\cos\theta=220\times91.77\times\cos11.33=19795.95W$

9-26 已知题 9-26 图中

$$u(t)=20\cos(10^3t+75°)\ V$$

$$i(t)=\sqrt{2}\sin(10^3t+120°)\ A$$

N_0 中无独立源。求 N_0 吸收的复功率和输入阻抗 Z_i。

$\dot{U}=10\sqrt{2}\angle75°$ V $\dot{I}=1\angle-60°$ A

$\dot{U}_L=\dot{I}\cdot j\omega L=1\angle30°$ V

$\dot{U}_R=\dot{I}\cdot R=2\angle-60°$ V

$\dot{U}_N+\dot{U}_R-\dot{U}+\dot{U}_L=0$

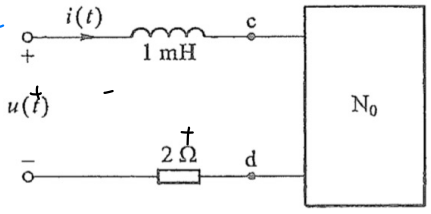
$\Rightarrow\dot{U}_N=\frac{-12+5\sqrt{3}}{2}+j\frac{9+12\sqrt{3}}{2}$

$\bar{S}=\dot{U}_N\cdot\dot{I}^*=-12+j9$ VA $Z_i=\frac{\dot{U}_N}{\dot{I}}=-12+j9\ \Omega$

$Z_{ab}=\frac{\dot{U}}{\dot{I}}=10+j10\ \Omega$

$Z_i=Z_{ab}-R-j\omega L=8+j9\ \Omega$

$\bar{S}=\dot{I}^2Z_i=8+j9$ VA



题 9-26 图

9-27 已知题 9-27 图中 $\dot{U}_s=100\angle90°$ V，$\dot{I}_s=5\angle0°$ A。求当 Z_L 获最大功率时各独立源发出的复功率。

外接阻法：$Z_{eq}=20\,//-j10\,//j20$

$=10-j10\ \Omega$

$\Rightarrow Z_L=10-j10\ \Omega$

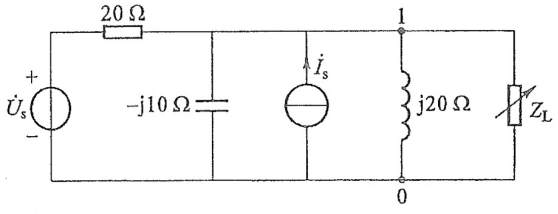
$(\frac{1}{20}+\frac{1}{-j10}+\frac{1}{j20}+\frac{1}{10-j10})\dot{U}_1=\frac{\dot{U}_s}{20}+\dot{I}_s$

$\Rightarrow\dot{U}_1=50$ V $50-j50$

$\bar{S}_I=\dot{U}_1\cdot\dot{I}_s^*=250$ VA $250+j250$ VA

$\dot{I}_u=\frac{\dot{U}_s-\dot{U}_1}{20}=2.5+j5$ A

$\bar{S}_u=\dot{U}_s\cdot\dot{I}_u^*=500-j250$ VA $500-j250$ VA



题 9-27 图